Exploring the Frontier: Generative and Diffusion Models in AI

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Abstract

This report presents an in-depth exploration of generative and diffusion models in artificial intelligence (AI), highlighting their transformative impact and emerging potential. These models, especially diffusion models, mark a significant progression in AI, moving from mere data analysis to the creation of novel, original data. They excel in generating high-quality, realistic outputs, akin to human creativity and thought processes. The core of this report includes a thorough examination of generative models, such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), along with an in-depth analysis of the principles and mechanisms underlying Denoising Diffusion Probabilistic Models. Furthermore, the report showcases the diverse applications of these models across various fields such as art, medicine, and entertainment, highlighting their versatility and wide-ranging impact.

1 Introduction

Generative models are a major advancement in artificial intelligence. Unlike standard algorithms that mainly analyze data, these models could create new data, mimicking the way humans think and create. A standout group within these are the diffusion models, known for their ability to produce outputs that are both high-quality and realistic. This development in AI results in a shift where machines not only learn from existing data but also generate their own outputs, ranging from artistic works to innovative solutions for various challenges.

2 Exploring Generative Models

Generative models in AI are algorithms designed to create new data instances that resemble a given dataset. They are completely different from discriminative models, which focus on categorizing or making predictions based on inputs. Generative models are able to capture and learn the distribution of the data, enabling them to produce new instances that are similar to the training data. Examples include Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs). GANs involve two neural networks—the generator and the discriminator—working in tandem in which the generator creates data and the discriminator evaluates it. VAEs, on the other hand, mainly work on encoding data into a lower-dimensional space and then reconstructing it back to the original form.

3 The Denoising Diffusion Probabilistic Models

3.1 Diffusion process

Given a data distribution $x_0 \sim q(x_0)$, define a forward diffusion process q which produces noised object x_t by adding Gaussian noise to x_{t-1} at time t with a varying variance $\beta_t \in (0, 1)$ as follows[1]:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I) \tag{1}$$

Given a sufficient large T and a well-behaved schedule of β_t, the resulting x_T closely approximates a standard Gaussian distribution $\mathcal{N}(0, 1)$. Then, define a reverse diffusion process p that denoising the object x_t to x_{t-1} by approximate the entire data distribution $p(x_{t-1}|x_t)$ using a neural network with parameters θ as follows[1]:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \tag{2}$$

3.2 Maximum Likelihood Estimation and Evidence Lower Bound

If we possess the precise reverse distribution $Pr(x_{t-1}|x_t)$, we can sample $x_T \sim \mathcal{N}(0, I)$ and run the diffusion process in reverse to generate an object from $Pr(x_0)$. In order to train a neural network with parameters θ that approximates the reverse distribution, we need to find the loss function of the diffusion model. Therefore, we employ log Maximum Likelihood Estimation (MLE) that maximize the $p(x_0)$ for all observed objects x_0:

$$\max_\theta log\ Pr(x_0) \tag{3}$$

By marginalizing out the noised objects $x_1, x_2, ..., x_T$, we can express the log-likelihood of x_0 as an Evidence Lower Bound (ELBO) in relation to the p and q functions, as outlined below:

$$\begin{aligned}
log\ p(x_0) &= log \int p(x_0, x_{1:T}) dx_{1:T} \\
&= log \int \frac{p(x_0, x_{1:T}) q(x_{1:T}|x_0)}{q(x_{1:T}|x_0)} dx_{1:T} \\
&= log\ \mathbb{E}_{q(x_{1:T}|x_0)} \left[\frac{p(x_0, x_{1:T})}{q(x_{1:T}|x_0)} \right] \\
&\leq \mathbb{E}_{q(x_{1:T}|x_0)} \left[log \frac{p(x_0, x_{1:T})}{q(x_{1:T}|x_0)} \right] = \mathbb{E}_{q(x_{1:T}|x_0)} \left[log \frac{p(x_T) \prod_{t=1}^T p_\theta(x_{t-1}|x_t)}{\prod_{t=1}^T q(x_t|x_{t-1})} \right]
\end{aligned} \tag{4}$$

The ELBO is a lower bound for the evidence, expressed here as the log-likelihood of the observed data x_0. Consequently, maximizing the ELBO becomes a proxy objective to optimize the original model. To facilitate comprehension and computation, we proceed to derive the ELBO, with detailed steps provided in Appendix A:

$$
\begin{aligned}
&ELBO = L_1 - \sum_{t=2}^{T} L_t - L_{T+1} \\
&L_1 := \mathbb{E}_{q(x_1|x_0)}[log\ p_\theta(x_0|x_1)] \\
&L_t := \mathbb{E}_{q(x_t,x_{t-1}|x_0)}\Big[D_{KL}(q(x_{t-1}|x_t,x_0)||p_\theta(x_{t-1}|x_t))\Big] \\
&L_{T+1} := D_{KL}(q(x_T|x_0)||p(x_T))
\end{aligned}
\tag{5}
$$

Presently, we've accomplished the derivation of the ELBO, where each term is computed as an expectation involving at most one random variable at a time. This formulation unveils an elegant interpretation for each individual term[2]:

1. Reconstruction term(L_1): predicting the log probability of the original data sample given the first-step latent.

2. Denoising matching term(L_t): Use a neural network with parameter θ to learn the reverse diffusion process $p_\theta(x_{t-1}x_t)$ to approximate the ground-truth denoising process $q(x_{t-1}|x_t, x_0)$. Therefore, this term is minimized when the two distributions match as closely as possible, as measured by their KL Divergence.

3. Prior matching term(L_{T+1}): this term represents how close the distribution of the final noised object is to the standard Gaussian prior. If we learn a perfect model, it is equal to zero(two distributions are the same).

3.3 Evaluate the Evidence Lower Bound

To assess the model, it's necessary to compute each term in the equation above. L_0 can be approximated and optimized through a Monte Carlo estimate. In the case of images, if we assume each color component is partitioned into 256 bins, we calculate the probability $p(x_0|x_1)$ of landing into the correct bin[3].

Regarding L_{T+1}, it represents a KL divergence between two Gaussian distributions without any trainable parameters, which can be evaluated in closed form. Within the KL divergence, there are two distributions: $p(x_T)$ is a standard normal distribution $\mathcal{N}(0, 1)$; to determine $q(x_T|x_0)$, we need to find $q(x_t|x_0)$ for any timestep t. For simplification in calculation, we define α_t and $\bar{\alpha}_t$ as follows[1]:

$$
\begin{aligned}
\alpha_t &= 1 - \beta_t \\
\bar{\alpha}_t &= \prod_{s=1}^{t} \alpha_s
\end{aligned}
\tag{6}
$$

Following this, we can express the forward diffusion process in terms of α rather than β:

$$
q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1-\alpha_t)I) \tag{7}
$$

Recall the reparameterization trick, which allows us to express a random variable as a deterministic function of a noise variable:

$$\begin{aligned} &x \sim \mathcal{N}(x; \mu, \sigma^2) \\ &x = \mu + \sigma\epsilon, \quad \epsilon \sim \mathcal{N}(0,1) \end{aligned} \tag{8}$$

We utilize the reparameterization trick on $x_t \sim q(x_t|x_{t-1})$ and obtain a formula for x_t in terms of x_0 and noise $\epsilon_t \sim \mathcal{N}(0,1)$:

$$\begin{aligned} x_t &= \sqrt{\alpha_t}x_{t-1} + \sqrt{1-\alpha_t}\epsilon_{t-1} \\ &= \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{\alpha_t(1-\alpha_{t-1})}\epsilon_{t-2} + \sqrt{1-\alpha_t}\epsilon_{t-1} \\ &= \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{\sqrt{\alpha_t-\alpha_t\alpha_{t-1}}^2 + \sqrt{1-\alpha_t}^2}\epsilon^*_{t-2} \\ &= \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\epsilon^*_{t-2} \\ &...... \\ &= \sqrt{\alpha_t\alpha_{t-1}...\alpha_0}x_0 + \sqrt{1-\alpha_t\alpha_{t-1}...\alpha_0}\epsilon^*_0 \\ &= \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon^*_0 \end{aligned} \tag{9}$$

From step 3 to step 4 in the equation above, we leverage the property that the sum of two independent Gaussian random variables yields a Gaussian distribution with the mean is the sum of the two means, and the variance is the sum of the two variances. Consequently, we obtain the distribution $q(x_t|x_0)$ for any timestep t:

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I) \tag{10}$$

The L_t term encompasses two distributions within the KL-divergence: $p_\theta(x_{t-1}x_t)$ and $q(x_{t-1}|x_t, x_0)$. The neural network approximates $p_\theta(x_{t-1}x_t)$, and we determine $q(x_{t-1}|x_t, x_0)$ by applying Bayes' theorem, with detailed calculations provided in Appendix B:

$$\begin{aligned} q(x_{t-1}|x_t, x_0) &= \frac{q(x_t|x_{t-1}, x_0)q(x_{t-1}|x_0)}{q(x_t|x_0)} \\ &= \frac{q(x_t|x_{t-1})q(x_{t-1}|x_0)}{q(x_t|x_0)} \\ &= \frac{\mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1-\alpha_t)I)\mathcal{N}(x_t; \sqrt{\bar{\alpha}_{t-1}}x_0, (1-\bar{\alpha}_{t-1})I)}{\mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I)} \\ &\propto \mathcal{N}(x_{t-1}; \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})x_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)x_0}{1-\bar{\alpha}_t}, \frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}I) \end{aligned} \tag{11}$$

To evaluate the KL divergence between $p_\theta(x_{t-1}x_t)$ and $q(x_{t-1}|x_t, x_0)$, recall the KL divergence between two normal distribution(d is the dimension of data):

$$D_{KL}(\mathcal{N}(x; \mu_x, \Sigma_x)||\mathcal{N}(y; \mu_y, \Sigma_y)) = \frac{1}{2}\left[log\,\frac{|\Sigma_y|}{\Sigma_x} - d + tr(\Sigma_y^{-1}\Sigma_x) + (\mu_y-\mu_x)^T\Sigma_y^{-1}(\mu_y-\mu_x)\right] \tag{12}$$

To match the approximate reverse process $p_\theta(x_{t1}|x_t)$ with the ground-truth denoising process $q(x_{t1}|x_t, x0)$, we represent the approximate process as a Gaussian with the same variance as the ground-truth process, since the variance of the ground-truth denoising process is solely determined by α and all α terms are fixed at each timestep. From the above discussion, minimizing L_t can maximize the ELBO to optimize the model. Consequently, we can plugin the distributions of the two denoising processes into the equation above:

$$
\begin{aligned}
& arg\min_\theta D_{KL}(q(x_{t-1}|x^t, x_0)||p_\theta(x_{t-1}|x_t)) \\
= & arg\min_\theta D_{KL}(\mathcal{N}(x_{t-1}; \mu_q, \Sigma)||\mathcal{N}(x_{t-1}; \mu_\theta, \Sigma)) \\
= & arg\min_\theta \frac{1}{2}\left[log\ \frac{|\Sigma|}{\Sigma} - d + tr(\Sigma^{-1}\Sigma) + (\mu_\theta - \mu_q)^T\Sigma^{-1}(\mu_\theta - \mu_q)\right] \\
= & arg\min_\theta \frac{1}{2}\left[log\ 1 - d + d + (\mu_\theta - \mu_q)^T\Sigma^{-1}(\mu_\theta - \mu_q)\right] \\
= & arg\min_\theta \frac{1}{2}\left[(\mu_\theta - \mu_q)^T\Sigma^{-1}(\mu_\theta - \mu_q)\right] \\
= & arg\min_\theta \frac{1}{2\sigma^2}\left[||\mu_\theta - \mu_q||_2^2\right]
\end{aligned} \tag{13}
$$

3.4 Loss function

After the above math derivation, the problem of optimizing the diffusion model reduces from maximizing the log-likelihood of observed data x_0 to minimizing the difference between the predicted mean and actual mean of denoising distributions. We can directly predict μ_θ using a neural network and employ gradient descent with the mean squared error between the predicted μ_θ and the μ of the distribution $q(x_{t-1}|x_t, x_0)$ in equation (11). However, for further model simplification, we can conduct additional derivations. Consider the sampling of x_t from the distribution $q(x_t|x_0)$ and rewrite the equation into a formula for x_0 in terms of x_t and noise ϵ:

$$
\begin{aligned}
x_t &= \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon_0 \\
x_0 &= \frac{x_t - \sqrt{1-\bar{\alpha}_t}\epsilon_0}{\sqrt{\bar{\alpha}_t}} \\
x_0 &= \frac{1}{\sqrt{\bar{\alpha}_t}}x_t - \sqrt{\frac{1-\bar{\alpha}_t}{\bar{\alpha}_t}}\epsilon_0
\end{aligned} \tag{14}
$$

Substitute the x_0 equation above into the μ of the distribution $q(x_{t-1}|x_t, x_0)$ in equation (11), and we obtain a μ that does not depend on x_0:

$$
\begin{aligned}
\mu_q &= \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})x_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)x_0}{1-\bar{\alpha}_t} \\
&= \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t}x_0 \\
&= \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \left(\frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t}\right)\left(\frac{1}{\sqrt{\bar{\alpha}_t}}x_t - \sqrt{\frac{1-\bar{\alpha}_t}{\bar{\alpha}_t}}\epsilon_0\right) \\
&= \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{1-\alpha_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}}x_t - \frac{1-\alpha_t}{\sqrt{\alpha_t(1-\bar{\alpha}_t)}}\epsilon_0 \\
&= \frac{1}{\sqrt{\alpha_t}}x_t - \frac{1-\alpha_t}{\sqrt{\alpha_t(1-\bar{\alpha}_t)}}\epsilon_0
\end{aligned}
\tag{15}
$$

We can express the mean μ_θ of the approximate reverse diffusion process $p_\theta(x_{t-1}|x_t)$ in a similar form:

$$
\begin{aligned}
p_\theta(x_{t-1}|x_t) &= \mathcal{N}(x_{t-1}; \mu_\theta, \Sigma) \\
\mu_\theta &= \frac{1}{\sqrt{\alpha_t}}x_t - \frac{1-\alpha_t}{\sqrt{\alpha_t(1-\bar{\alpha}_t)}}\epsilon_\theta(x_t, t)
\end{aligned}
\tag{16}
$$

Then, we can further derive the equation (13):

$$
\begin{aligned}
& arg \min_\theta D_{KL}(q(x_{t-1}|x^t, x_0)||p_\theta(x_{t-1}|x_t)) \\
=\ & arg \min_\theta D_{KL}(\mathcal{N}(x_{t-1}; \mu_q, \Sigma)||\mathcal{N}(x_{t-1}; \mu_\theta, \Sigma)) \\
=\ & arg \min_\theta \frac{1}{2\sigma^2}\left[||\mu_\theta - \mu_q||_2^2\right] \\
=\ & arg \min_\theta \frac{1}{2\sigma^2}\left[||\left(\frac{1}{\sqrt{\alpha_t}}x_t - \frac{1-\alpha_t}{\sqrt{\alpha_t(1-\bar{\alpha}_t)}}\epsilon_\theta(x_t, t)\right) - \left(\frac{1}{\sqrt{\alpha_t}}x_t - \frac{1-\alpha_t}{\sqrt{\alpha_t(1-\bar{\alpha}_t)}}\epsilon_0\right)||_2^2\right] \\
=\ & arg \min_\theta \frac{1}{2\sigma^2}\left[||\left(\frac{1-\alpha_t}{\sqrt{\alpha_t(1-\bar{\alpha}_t)}}\right)(\epsilon_\theta(x_t, t) - \epsilon_0)||_2^2\right] \\
=\ & arg \min_\theta \frac{(1-\alpha_t)^2}{2\sigma^2\alpha_t(1-\bar{\alpha}_t)}\left[||\epsilon_\theta(x_t, t) - \epsilon_0||_2^2\right]
\end{aligned}
\tag{17}
$$

As a result, we obtain a straightforward loss function $||\epsilon_\theta(x_t, t) - \epsilon_0||_2^2$ for the diffusion model. This implies that the network is focused on predicting only the noise ϵ added to the object.

3.5 Algorithm

This section will include pseudocode outlining our diffusion model algorithm. We begin with the training process of the neural network:

Algorithm 1 Training diffusion model

initialize neural network NN_θ with parameter θ
initialize α_t for $t \in [1, T]$
while NN is not converged **do**
    $x_0 \sim p(x_0)$
    $t \sim \{1, 2, ..., T\}$
    $\epsilon \sim \mathcal{N}(0, 1)$
    $x_t \leftarrow \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_0$
    $\epsilon_\theta \leftarrow NN_\theta(x_t, t)$
    loss $= ||\epsilon - \epsilon_\theta||_2^2$
    takes gradient descent on loss within NN_θ
end while

Before training the neural network, we need to initialize it with its initial parameters θ. Additionally, we must initialize all α_t values for each timestep t as $\alpha_t = 1 - \beta_t$, where β_t follows a noise schedule ranging from zero to one. In the training process, we start by randomly sampling an object x_0 from the dataset $p(x_0)$. Then, we sample a timestep t and noise from the standard normal distribution $\mathcal{N}(0, 1)$. Following that, we use the neural network to predict the noise at timestep t, using the noised object x_t as input. Finally, we optimize our model through gradient descent on the loss function.

4 Applications and Case Studies

Explore various applications of generative and diffusion models in different industries, including art, medicine, and entertainment. Present case studies where these models have been successfully implemented.

4.1 Art

- Digital Art Creation: Refik Anadol's generative AI-based artworks utilize data from diverse sources, such as real-time weather data and historic architecture, to create digital art that changes people's perspectives of their surroundings [4].

- The Next Rembrandt: A collaboration involving data scientists, art historians, and Microsoft, this project used generative AI to create a painting in the style of Rembrandt, showcasing the AI's ability to learn and replicate an artist's style [5].

4.2 Medicine

- Drug Discovery and Development: Generative models are increasingly used in the pharmaceutical industry for predicting drug interactions and accelerating the discovery of new drugs.

- Medical Imaging Enhancement: Diffusion models are applied to enhance the quality and resolution of medical images like MRIs and X-rays, aiding in more accurate diagnoses.

4.3 Entertainment

- Video Game Design: Generative AI is used in gaming to create dynamic and realistic game environments, contributing to more immersive gaming experiences.

- Film Industry: In movies, generative AI is employed for creating realistic special effects and can be used to generate entire scenes or characters.

5 Challenges and Limitations

Developing and deploying generative and diffusion models come with several challenges and limitations, including ethical considerations. One major challenge is ensuring fairness and avoiding biases in these models. Since they learn from data, any inherent biases in the training data can lead to biased outputs, potentially reinforcing stereotypes or unfair representations. Another significant challenge is the risk of misuse, especially in generating misleading or harmful content, which raises serious ethical concerns.

There are also technical limitations, such as the quality and diversity of the generated outputs. Sometimes, these models might produce unrealistic or irrelevant results, or their outputs might lack the subtlety and depth of human-created content. Moreover, the computational resources required for training and operating these models are substantial, making them less accessible for smaller organizations or individuals.

Ensuring the responsible use of these technologies involves addressing these ethical and technical challenges, and balancing innovation with caution to prevent misuse and negative societal impacts.

6 Future Prospects

The future of generative and diffusion models in AI is poised for significant advancements and broader impacts across various fields. As these models continue to evolve, we anticipate several key developments:

- **Increased Realism and Detail in Generated Content:** Future models are expected to produce even more realistic and intricate outputs, blurring the line between AI-generated and human-created content. This advancement could revolutionize fields like digital art, entertainment, and virtual reality.

- **Enhanced Personalization and Accessibility:** As these models become more sophisticated, they will likely offer greater personalization in applications such as content creation, enabling users to generate highly specific and tailored results with minimal input.

- **Expansion into New Domains:** We foresee the application of generative and diffusion models expanding into new domains, including more intricate work in scientific research, environmental modeling, and even in areas like urban planning and architecture.
- **Improvements in Efficiency and Sustainability:** Future models are likely to become more efficient, requiring less computational power and thus becoming more sustainable and accessible to a wider range of users and applications.
- **Ethical and Regulatory Developments:** As these technologies advance, there will be an increased focus on ethical considerations and the development of regulations to ensure responsible use, particularly in areas like deepfake detection and the creation of digital media.

These advancements will not only push the boundaries of what's possible with AI but will also pose new challenges and ethical considerations, necessitating a balanced approach to innovation and application.

7 Conclusion

In brief, generative and diffusion models are truly remarkable in the field of AI. Unlike traditional machine learning models that just analyze and make sense of data, these models can actually create new and original data. They're not just limited to understanding what they're fed; they can innovate and come up with fresh, creative outputs. As technology keeps improving, these models are expected to become increasingly important across various fields, leading the way in innovation and creative solutions.

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Appendix A Evidence Lower Bound (ELBO) derivation

Recall that the ELBO of the diffusion model is:

$$ELBO = \mathbb{E}_{q(x_{1:T}|x_0)}\left[log\frac{p(x_0, x_{1:T})}{q(x_{1:T}|x_0)}\right] \tag{18}$$

We first derive the log term inside the expectation:

$$\begin{aligned}
log\frac{p(x_0, x_{1:T})}{q(x_{1:T}|x_0)} &= log\frac{p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)}{\prod_{t=1}^{T} q(x_t|x_{t-1})} \\
&= log\frac{p(x_T)p_\theta(x_0|x_1)\prod_{t=2}^{T} p_\theta(x_{t-1}|x_t)}{q(x_1|x_0)\prod_{t=2}^{T} q(x_t|x_{t-1})} \\
&= log\frac{p(x_T)p_\theta(x_0|x_1)\prod_{t=2}^{T} p_\theta(x_{t-1}|x_t)}{q(x_1|x_0)\prod_{t=2}^{T} q(x_t|x_{t-1}, x_0)} \\
&= log\frac{p(x_T)p_\theta(x_0|x_1)}{q(x_1|x_0)} + log\prod_{t=2}^{T}\frac{p_\theta(x_{t-1}|x_t)}{q(x_t|x_{t-1}, x_0)} \\
&= log\frac{p(x_T)p_\theta(x_0|x_1)}{q(x_1|x_0)} + log\prod_{t=2}^{T}\frac{p_\theta(x_{t-1}|x_t)}{\frac{q(x_{t-1}|x_t, x_0)q(x_t|x_0)}{q(x_{t-1}|x_0)}} \\
&= log\frac{p(x_T)p_\theta(x_0|x_1)}{q(x_1|x_0)} + log\frac{q(x_1|x_0)}{q(x_T|x_0)} + log\prod_{t=2}^{T}\frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)} \\
&= log\frac{p(x_T)p_\theta(x_0|x_1)}{q(x_T|x_0)} + log\prod_{t=2}^{T}\frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)} \\
&= log\ p_\theta(x_0|x_1) + log\frac{p(x_T)}{q(x_T|x_0)} + \sum_{t=2}^{T} log\frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)}
\end{aligned} \tag{19}$$

On the second line to the third line above, utilize a trick $q(x_t|x_{t-1}) = q(x_t|x_{t-1}, x_0)$. Then, apply the log term inside the expectation:

$$
\begin{aligned}
&\mathbb{E}_{q(x_{1:T}|x_0)}\left[log\frac{p(x_0,x_{1:T})}{q(x_{1:T}|x_0)}\right] \\
=&\mathbb{E}_{q(x_{1:T}|x_0)}\left[log\ p_\theta(x_0|x_1)+log\frac{p(x_T)}{q(x_T|x_0)}+\sum_{t=2}^{T}log\frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t,x_0)}\right] \\
=&\mathbb{E}_{q(x_{1:T}|x_0)}[log\ p_\theta(x_0|x_1)]+\mathbb{E}_{q(x_{1:T}|x_0)}\left[log\frac{p(x_T)}{q(x_T|x_0)}\right] \\
&+\sum_{t=2}^{T}\mathbb{E}_{q(x_{1:T}|x_0)}\left[log\frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t,x_0)}\right] \\
=&\mathbb{E}_{q(x_1|x_0)}[log\ p_\theta(x_0|x_1)]+\mathbb{E}_{q(x_T|x_0)}\left[log\frac{p(x_T)}{q(x_T|x_0)}\right] \\
&+\sum_{t=2}^{T}\mathbb{E}_{q(x_{t-1},x_t|x_0)}\left[log\frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t,x_0)}\right] \\
=&\mathbb{E}_{q(x_1|x_0)}[log\ p_\theta(x_0|x_1)]-D_{KL}(q(x_T|x_0)||p(x_T)) \\
&-\sum_{t=2}^{T}\mathbb{E}_{q(x_{t-1},x_t|x_0)}\left[D_{KL}(q(x_{t-1}|x_t,x_0)||p_\theta(x_{t-1}|x_t))\right]
\end{aligned}
\tag{20}
$$

Therefore, we have

$$
\begin{aligned}
ELBO = &\ \mathbb{E}_{q(x_1|x_0)}[log\ p_\theta(x_0|x_1)] \\
&-\ \sum_{t=2}^{T}\mathbb{E}_{q(x_t,x_{t-1}|x_0)}\left[D_{KL}(q(x_{t-1}|x_t,x_0)||p_\theta(x_{t-1}|x_t))\right] \\
&-\ D_{KL}(q(x_T|x_0)||p(x_T))
\end{aligned}
\tag{21}
$$

Appendix B The ground-truth denoising distribution derivation

$$
\begin{aligned}
q(x_{t-1}|x_t,x_0) &= \frac{q(x_t|x_{t-1},x_0)q(x_{t-1}|x_0)}{q(x_t|x_0)} \\
&= \frac{q(x_t|x_{t-1})q(x_{t-1}|x_0)}{q(x_t|x_0)} \\
&= \frac{\mathcal{N}(x_t;\sqrt{\alpha_t}x_{t-1},(1-\alpha_t)I)\mathcal{N}(x_t;\sqrt{\bar{\alpha}_{t-1}}x_0,(1-\bar{\alpha}_{t-1})I)}{\mathcal{N}(x_t;\sqrt{\bar{\alpha}_t}x_0,(1-\bar{\alpha}_t)I)} \\
&\propto exp\Big\{-\Big[\frac{(x_t-\sqrt{a_t}x_{t-1})_2}{2(1-\alpha_t)}+\frac{(x_{t-1}-\sqrt{\bar{\alpha}_{t-1}}x_0)^2}{2(1-\bar{\alpha}_{t-1})}-\frac{(x_t-\sqrt{\bar{\alpha}_t}x_0)^2}{2(1-\bar{\alpha}_t)}\Big]\Big\} \\
&= exp\Big\{-\Big[\frac{-2\sqrt{\alpha_t}x_tx_{t-1}+\alpha_tx_{t-1}^2}{2(1-\alpha_t)}+\frac{x_{t-1}^2-2\sqrt{\bar{\alpha}_{t-1}}x_{t-1}x_0}{2(1-\bar{\alpha}_{t-1})}+C(x_t,x_0)\Big]\Big\} \\
&\propto exp\Big\{-\frac{1}{2}\Big[-\frac{2\sqrt{\alpha_t}x_t}{1-\alpha_t}x_{t-1}+\frac{\alpha_t}{1-\alpha_t}x_{t-1}^2+\frac{1}{1-\bar{\alpha}_{t-1}}x_{t-1}^2-\frac{2\sqrt{\bar{\alpha}_{t-1}}x_0}{1-\bar{\alpha}_{t-1}}x_{t-1}\Big]\Big\} \\
&= exp\Big\{-\frac{1}{2}\Big[(\frac{\alpha_t}{1-\alpha_t}+\frac{1}{1-\bar{\alpha}_{t-1}})x_{t-1}^2-2(\frac{\sqrt{\alpha_t}x_t}{1-\alpha_t}+\frac{\sqrt{\bar{\alpha}_{t-1}}x_0}{1-\bar{\alpha}_{t-1}})x_{t-1}\Big]\Big\} \\
&= exp\Big\{-\frac{1}{2}\Big[\frac{1-\bar{\alpha}_t}{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}x_{t-1}^2-2(\frac{\sqrt{\alpha_t}x_t}{1-\alpha_t}+\frac{\sqrt{\bar{\alpha}_{t-1}}x_0}{1-\bar{\alpha}_{t-1}})x_{t-1}\Big]\Big\} \\
&= exp\Big\{-\frac{1}{2}(\frac{1-\bar{\alpha}_t}{(1-\alpha_t)(1-\bar{\alpha}_{t-1})})\Big[x_{t-1}^2-2\frac{(\frac{\sqrt{\alpha_t}x_t}{1-\alpha_t}+\frac{\sqrt{\bar{\alpha}_{t-1}}x_0}{1-\bar{\alpha}_{t-1}})}{\frac{1-\bar{\alpha}_t}{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}}x_{t-1}\Big]\Big\} \\
&= exp\Big\{-\frac{1}{2}(\frac{1}{\frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}})\Big[x_{t-1}^2-2\frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})x_t+\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)x_0}{1-\bar{\alpha}_t}x_{t-1}\Big]\Big\} \\
&\propto \mathcal{N}(x_{t-1};\frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})x_t+\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)x_0}{1-\bar{\alpha}_t},\frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}I)
\end{aligned} \tag{22}
$$